

Relativistic Time Dilation & Coordinate Phase Integration

Theoretical Mathematics of the Time Dilation DAW Engine

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Abstract

This document details the mathematical framework governing relativistic coordinate time dilation, Lorentz velocity transforms, gravitational redshift event horizons, and digital audio phase integration in **Time Dilation DAW (v4.0)**.

1 Introduction to Relativistic Coordinate Time (γ)

In standard Digital Audio Workstations (DAWs), digital time progresses linearly at a constant sample rate f_s :

$$t(n) = \frac{n}{f_s} \quad (1)$$

In **Time Dilation DAW**, coordinate time is modulated by a dynamic relativistic factor $\gamma(t) \in [-16.0, 16.0]$, representing localized gravitational or velocity time dilation.

2 Velocity Time Dilation (Special Relativity)

Under Einstein's Special Theory of Relativity, coordinate time dilation for a system moving at relative velocity v with respect to speed of light c is defined by the Lorentz factor:

$$\gamma(v) = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (2)$$

For $v \rightarrow c$, $\gamma(v) \rightarrow \infty$, causing asymptotic time dilation. In our engine, γ is clamped to $[-16.0, 16.0]$ for real-time digital stability.

3 Gravitational Time Dilation & Event Horizon Redshift

Near a massive gravitational body of Schwarzschild radius $r_s = \frac{2GM}{c^2}$, time dilation at radius r follows:

$$\gamma(r) = \frac{1}{\sqrt{1 - \frac{r_s}{r}}} = \frac{1}{\text{redshift}} \quad (3)$$

As $r \rightarrow r_s$ (approaching the Event Horizon), redshift $\rightarrow 0$, causing time stasis ($\gamma \rightarrow 0.0$).

4 Coordinate Time Accumulation & Phase Integration

The cumulative relativistic coordinate time $\tau(t)$ is obtained by integrating $\gamma(t')$ over physical timeline t :

$$\tau(t) = \int_0^t \gamma(t') dt' \quad (4)$$

In discrete time at sample rate f_s , the per-sample phase increment $\Delta\phi(n)$ for an oscillator at frequency f is:

$$\Delta\phi(n) = \frac{2\pi \cdot f \cdot |\gamma(n)|}{f_s} \quad (5)$$

When $\gamma(n) < 0$, the phase steps backward ($\Delta\phi < 0$), executing authentic *Retrograde Time Reversal*.